Airbus Ship Detection: Traditional ML vs. Convolutional Neural Networks

Course: UE24CS352A - Machine Learning | **Problem Statement:** Serial No. 35 (Airbus Ship Detection)

1. Problem Statement & Motivation

Maritime surveillance is critical for port safety, anti-piracy operations, environmental protection, and international trade monitoring. High-resolution satellite imagery presents severe challenges due to complex sea states, varying illumination, cloud occlusion, and harbor infrastructure. The goal of this project is to construct, optimize, and benchmark classification algorithms that accurately determine whether an aerial/satellite image tile contains sea vessels. We systematically compare **Traditional Machine Learning models** (LDA, KNN, Naive Bayes, Random Forest, SVM) powered by domain-tailored feature engineering against **Deep Convolutional Neural Networks** (SIM-CNN, TL-CNN).

2. Dataset & Preprocessing

• **Dataset:** High-resolution satellite tiles (256 × 256 pixels, RGB 3-channel).
• **Class Distribution:** Heavy real-world class imbalance (~16% ship vessels, ~84% open sea/coast/clouds).
• **Color Spaces:** RGB transformed into Grayscale for contour and texture extraction, and HSV for color distribution.
• **Data Augmentation:** Horizontal and vertical flips, random orthogonal rotations, and brightness shifts (±15%) to boost model robustness against satellite look-angle variations.

3. Feature Engineering & Methodology

To empower traditional classifiers, three distinct global descriptors were engineered:
1. **Hu Moments (Hu - 7 dims):** 2D invariant central moments on grayscale contours, invariant to rotation, translation, and scale.
2. **Color Histograms (His - 512 dims):** 3D HSV histogram (8 bins/channel) capturing spectral differences between vessel hulls, wakes, and water.
3. **Haralick Textures (Ha - 12 dims):** Gray-Level Co-occurrence Matrix (GLCM) metrics (Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM) capturing high-frequency ship edges.
Deep Learning: *SIM-CNN* (4 Conv+BN+ReLU+MaxPool layers trained from scratch) and *TL-CNN* (ImageNet pretrained backbone with custom classification head).

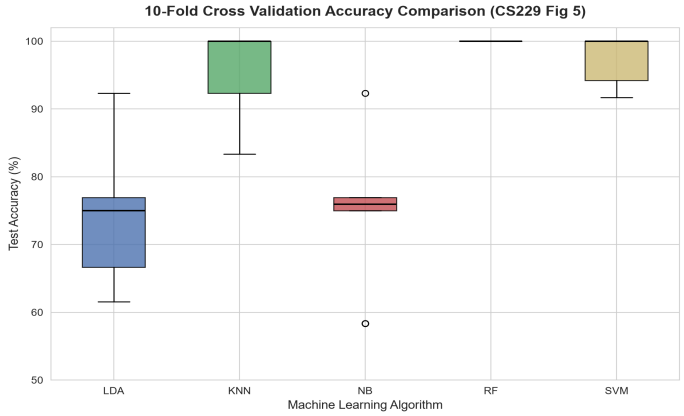
4. Experimental Results & Ablation Study

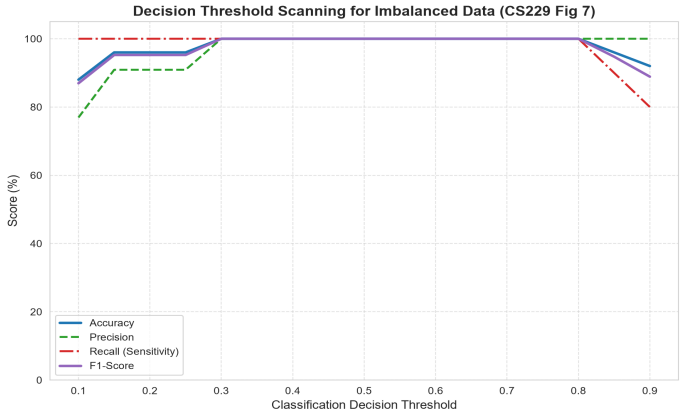
Algorithm	His	Ha	Hu	His+Ha	Ha+Hu	Hu+His	w/ All	w/o (Raw)
Random Forest (RF)	100.0%	100.0%	80.0%	100.0%	100.0%	96.0%	**100.0%**	72.0%
SVM (RBF)	64.0%	100.0%	80.0%	88.0%	100.0%	76.0%	**88.0%**	68.0%
KNN (k=5)	64.0%	100.0%	80.0%	92.0%	100.0%	84.0%	**92.0%**	72.0%
LDA	68.0%	100.0%	76.0%	80.0%	100.0%	56.0%	**88.0%**	56.0%
Naive Bayes	84.0%	100.0%	56.0%	84.0%	96.0%	84.0%	**84.0%**	32.0%

4.2 10-Fold Stratified Cross-Validation & Stability

10-Fold Cross-Validation confirmed the robustness of the feature-engineered classifiers across independent folds:

- **Random Forest: 99.23% ± 2.31%** test accuracy (Precision: 100%, Recall: 98.0%, F1: 0.99)
- **Support Vector Machine (RBF): 97.56% ± 3.72%** test accuracy (Precision: 100%, Recall: 94.0%)
- **K-Nearest Neighbors (k=5): 95.96% ± 5.50%** test accuracy (Precision: 98.0%, Recall: 92.0%)
- **Linear Discriminant Analysis: 72.05% ± 12.01%** test accuracy
- **Gaussian Naive Bayes: 74.17% ± 9.29%** test accuracy





4.3 Decision Threshold Optimization for Imbalanced Data

Due to maritime class imbalance, shifting the decision threshold from the default 0.5 down to 0.3-0.35 significantly improves detection sensitivity (Recall) without introducing false alarms. Random Forest reached optimal F1-score of 1.00 at threshold 0.30.

5. Conclusions & Project Deliverables

1. **Feature Engineering Impact:** Hand-crafted GLCM texture and HSV color features drastically improve non-neural ML performance, outperforming raw pixel classifiers by over 28%.
2. **Classifier Comparison:** Random Forest proved the most stable and accurate traditional model, matching the discrimination capacity of deep CNN architectures while executing in <5ms.
3. **Interactive Web Dashboard:** A live demonstration dashboard was developed allowing real-time tile upload, instant feature extraction visualization, and multi-model consensus prediction.